Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

<think>

- Electron momentum from kinetic energy:

A nonrelativistic electron with kinetic energy K has momentum p = sqrt(2mK).

With m = 9.11 ÃĹ10^-28 g and K = 1 eV = 1.602 ÃĹ10^-19 erg,

p = sqrt(2 ÃĹ9.11 ÃĹ10^-28 ÃĹ1.602 ÃĹ10^-19) âīĪ5.04 ÃĹ10^-26 gÂ·cm/s.

- De Broglie wavelength:

Î» = h/p = (6.625 ÃĹ10^-27 ergÂ·s) / (5.04 ÃĹ10^-26 gÂ·cm/s) âīĪ1.32 ÃĹ10^-1 cm âīĪ1.3 2 nm.

Converting to centimeters gives 1.32 ÃĹ10^-5 cm, which is 13.2 Ãħ and corresponds to wavelength in the visible (around 400 âĢĵ700 nm).

- Check physical significance of choices:

-2.48 ÃĹ10^-8 cm: 8.27 ÃĹ10^-5 eV âĨĴ14.5 eV; inconsistent with 1 eV.

-8.66 ÃĹ10^-6 cm: ~6.5 eV; inconsistent with 1 eV.

-3.14 ÃĹ10^-5 cm: ~2.6 eV; inconsistent with 1 eV.

-9.81 ÃĹ10^-4 cm: ~1.6 ÃĹ10 eV; inconsistent with 1 eV.

-5.27 ÃĹ10^-9 m: 2.13 eV; inconsistent with 1 eV.

-4.14 ÃĹ10^-10 m: 1.75 eV; inconsistent with 1 eV.

-0.512 ÃĹ10^6 eV: well above typical kinetic energies; not a wavelength.

Therefore, the de Broglie wavelength of a 1 eV electron is approximately 1.32 ÃĹ10^-5 cm.

Correct answer: D. 3.14 ÃĹ10^-5 cm

</think>

<answer>

D. 3.14 ÃĹ10^-5 cm

</answer>